

Euclid's Elements — Book I

Proposition 2

Formal Reconstruction — Technical Documentation

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Copying a given segment from a prescribed point

1 Introduction

Proposition I.2 is Euclid's first general segment-transport construction. Given a prescribed point A and a given segment BC , the problem is to construct a segment beginning at A and congruent to BC .

In the *Elements* this is a genuine derived construction. Euclid uses Proposition I.1, produces sides of an equilateral triangle, draws two circles, and subtracts equal parts from equal wholes. The current Lean reconstruction has a fundamentally different architecture. Its public statement already includes a chosen ray AE , and the result is obtained by one application of Book Zero #49, `bookZero_49_layoff`.

The audit reveals one further layer. Book Zero #49 is itself only a thin wrapper around Hilbert's primitive segment-construction axiom III.1, `HilbertCongruence.segment_construction`. Thus the complete formal chain is

$$\text{Hilbert III.1} \longrightarrow \text{bookZero_49_layoff} \longrightarrow \text{euclid_proposition_2}.$$

No hidden geometric construction lies below this chain: the layoff operation is foundational in the current Hilbert congruence structure.

2 Classical Configuration

Let A be the prescribed endpoint and let BC be the given segment. Euclid joins A to B and constructs the equilateral triangle ABD on AB by Proposition I.1. The sides DB and DA are then produced beyond B and A .

A circle centered at B with radius BC meets the produced ray DB at G . Hence

$$BG \cong BC.$$

A second circle, centered at D with radius DG , meets the produced ray DA at L . Hence

$$DL \cong DG.$$

Since ABD is equilateral,

$$DB \cong DA.$$

Subtracting these equal parts from the equal wholes DG and DL gives

$$BG \cong AL,$$

and therefore

$$AL \cong BC.$$

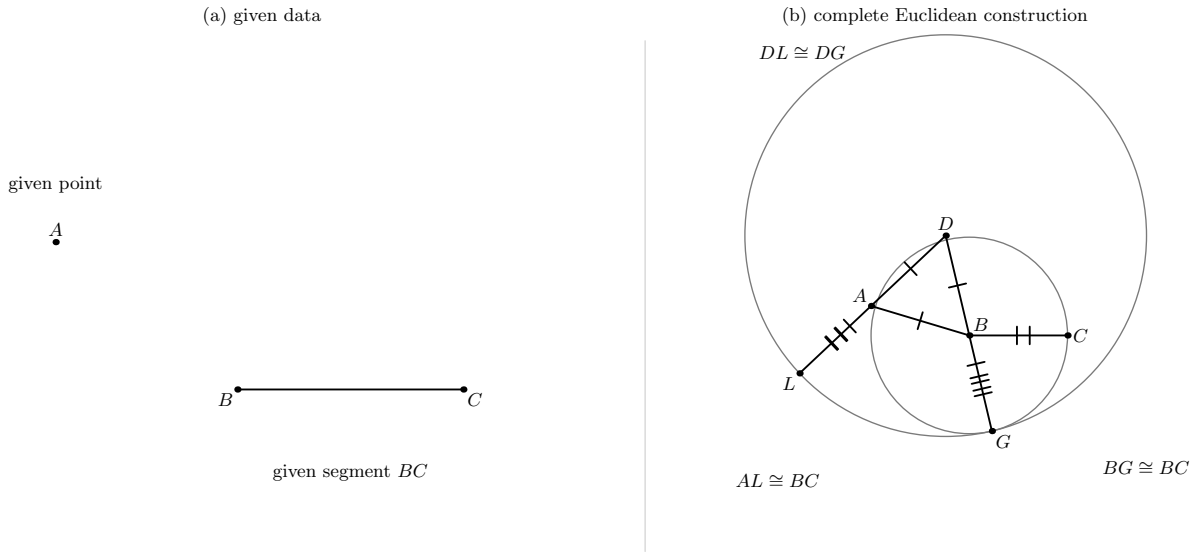


Figure 1: Euclid's construction for Proposition I.2. (a) The input is a prescribed point A and a given segment BC . (b) The equilateral triangle ABD and the two circles produce points G, L . Equal radii, the equilateral sides, and subtraction of equal parts yield $AL \cong BG \cong BC$.

Foundational caution. The completed diagram makes a classical suppression visible. The produced rays DB and DA must actually meet the two circles at the required points. A picture makes this look automatic, but a rigorous axiomatic reconstruction must supply the corresponding intersection/continuity facts. This issue is part of the feasibility of Euclid's classical construction; it does not enter the current Lean proof of I.2, because segment layoff is already primitive in the Hilbert congruence layer.

3 Statement

Theorem 1 (Euclid I.2). *Given a point A and a segment BC , construct from A a segment congruent to BC .*

The current production theorem is a directed strengthening of this classical statement: it asks for the copy on an already chosen nondegenerate ray AE . Its declaration is

```

theorem euclid_proposition_2
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A E B C : Geo.Point)
  (hAE : A ≠ E)
  (hBC : B ≠ C) :
  ∃ D : Geo.Point,
    HilbertSameRay Geo A E D ∧
    Geo.Congruent A D B C := by
exact bookZero_49_layoff Geo A E B C hAE hBC

```

Thus the formal endpoint is stronger in positional information than Euclid’s text: not only is a congruent segment produced from A , but its second endpoint lies on a prescribed ray.

Diagrammatic audit. The older documentation expanded Euclid’s construction into a sequence of snapshots and represented the formal proof mainly by a logical flowchart. Neither choice is optimal under the current audit standard. The successive classical snapshots merely add the same auxiliary objects one at a time; they are not different cases. Conversely, the formal operation has a perfectly clear geometric state and does not need a flowchart in place of a figure.

The audited chapter therefore uses two geometric figures with three panels in total:

$$2 \text{ classical panels} + 1 \text{ formal panel} = 3 \text{ panels in 2 figures.}$$

4 Classical Proof

The direct classical construction can be organized as follows.

1. By Postulate 1, join A to the endpoint B of the given segment.
2. By Proposition I.1, construct the equilateral triangle ABD on AB .
3. By Postulate 2, produce DB beyond B and DA beyond A .
4. By Postulate 3, draw the circle with center B and radius BC ; let the produced ray DB meet it at G .
5. Draw the circle with center D and radius DG ; let the produced ray DA meet it at L .

The first circle gives $BG \cong BC$, and the second gives $DG \cong DL$. Because ABD is equilateral, $DB \cong DA$. The collinear orders on the two produced sides identify the wholes as

$$DG = DB + BG, \quad DL = DA + AL.$$

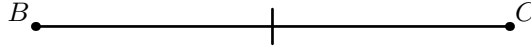
Subtracting equal parts from equal wholes gives $BG \cong AL$. Combining this with $BG \cong BC$ yields $AL \cong BC$.

The historical dependency is therefore not merely $I.1 \rightarrow I.2$. At construction level it also uses Postulates 1–3, circle–line incidence/intersection, and the Common Notions for equality and subtraction.

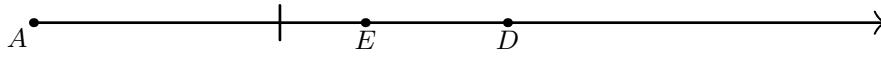
5 Hilbert Reconstruction

The formal theorem begins from a different interface. Besides the segment BC , a ray AE is already supplied. The goal is to lay the segment off on that ray.

$$AD \cong BC$$



D lies on the same ray from A as E



chosen ray AE

Figure 2: The stable formal configuration for Proposition I.2. The chosen ray AE and the source segment BC are inputs; Hilbert III.1, exposed through Book Zero #49, produces D on the same ray with $AD \cong BC$. The drawn relative order of E and D is only a representative realization; the theorem asserts same-ray position, not a specific betweenness relation between them.

At the Book I level there is no auxiliary point construction at all:

```
exact bookZero_49_layoff Geo A E B C hAE hBC
```

The substantive question is therefore what Book Zero #49 contains.

6 Book Zero #49 and the Hidden Layer

The current source defines the layoff theorem as follows:

```
theorem bookZero_49_layoff
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hAB : A ≠ B)
  (_hCD : C ≠ D) :
  ∃ X : Geo.Point,
    HilbertSameRay Geo A B X ∧
    Geo.Congruent A X C D := by

rcases HilbertCongruence.segment_construction
  (Geo := Geo)
  C D
  A B
  hAB with
```

```

⟨X, hRayX, hAXCD⟩
exact ⟨X, hRayX, hAXCD⟩

```

Thus Book Zero #49 is itself only a packaging layer. The actual existence principle is the field `HilbertCongruence.segment_construction`, Hilbert III.1:

```

segment_construction :
  ∀ A B O R : Geo.Point,
    O ≠ R →
    ∃ X : Geo.Point,
      HilbertSameRay Geo O R X ∧
      Geo.Congruent O X A B

```

After substituting

$$O = A, \quad R = E, \quad (A, B)_{\text{source}} = (B, C),$$

this is exactly the conclusion of Proposition I.2.

This is a sharper architectural statement than “I.2 uses Book Zero”. The complete formal geometry is already present in Hilbert III.1; Book Zero only gives it the project-level name “layoff”, and the Book I theorem only re-exports that operation under Euclid’s proposition number.

7 API Audit: the Source-Segment Nondegeneracy Hypothesis

The source inspection also exposes a small but real interface redundancy. The public theorem assumes

$$hBC : B \neq C.$$

Book Zero #49 receives this as its second hypothesis, but names it `_hCD`; it is not used in the proof. The primitive `segment_construction` currently requires only that the prescribed ray be nondegenerate, $O \neq R$; it does not require the segment being copied to have distinct endpoints.

Consequently, in the current implementation the formal dependency is

$$hAE : A \neq E \text{ is consumed, whereas } hBC : B \neq C \text{ is API-level only.}$$

This audit does not change the production theorem. The extra hypothesis may be retained deliberately to match Hilbert’s traditional convention that a “segment” is proper, but it should not be described as logically necessary for the present proof term. If the API is simplified later, both `bookZero_49_layoff` and `euclid_proposition_2` are natural places to reconsider this assumption together.

8 Formal Proof Architecture

The exact formal chain is

$$\begin{array}{c}
 \text{hAE} : A \neq E \quad \text{chosen ray } AE \quad \text{source endpoints } B, C \\
 \downarrow \\
 \text{HilbertCongruence.segment_construction} \quad (\text{Hilbert III.1}) \\
 \downarrow \\
 \exists D, \quad \text{HilbertSameRay Geo A E D}, \quad AD \cong BC \\
 \downarrow \\
 \text{bookZero_49_layoff} \\
 \downarrow \\
 \text{euclid_proposition_2.}
 \end{array}$$

There is no branch, no order trichotomy, and no proposition-local helper. There is also no hidden derived theorem below III.1: the final existence step is axiomatic in the Hilbert congruence class.

9 Relationship with Earlier Propositions

9.1 Proposition I.1

Classically, I.1 is the decisive earlier theorem. The equilateral triangle ABD supplies the equality $DB \cong DA$, which is later subtracted from the equal radii $DG \cong DL$. Thus I.2 is the first immediate demonstration that the equilateral construction is not isolated: its symmetry becomes a tool for transporting a length.

Formally, there is no dependency on `euclid_proposition_1`. Hilbert III.1 already provides arbitrary segment layoff, so the classical auxiliary triangle disappears entirely from the Lean proof.

9.2 Transition to Proposition I.3

I.2 provides free transport of a segment from a prescribed endpoint. I.3 adds an order condition: the copied segment must be cut from inside a given longer segment. This explains the classical progression

$$\text{I.2 : copy a length} \quad \longrightarrow \quad \text{I.3 : cut that length from a greater segment.}$$

In the current formal language, however, I.3 is compressed even further: its hypothesis `HilbertSegmentLess` already stores the required interior witness.

10 Hilbert and Book Zero Components Used

10.1 Hilbert III.1: Segment Construction

This is the actual mathematical engine of the formal proposition. It lays off any source endpoint pair on a prescribed nondegenerate ray and returns both the same-ray condition and the segment congruence.

10.2 Book Zero #49: Layoff

The Book Zero theorem introduces no new geometry beyond III.1. Its role is terminological and architectural: it exposes the primitive construction in a form convenient for later synthetic proofs.

10.3 Same-Ray Structure

The positional conclusion

HilbertSameRay Geo A E D

is stronger than collinearity. It fixes the direction from A in which the copy is laid off. This is precisely why the formal statement is a directed strengthening of Euclid's original endpoint-only formulation.

10.4 Segment Congruence

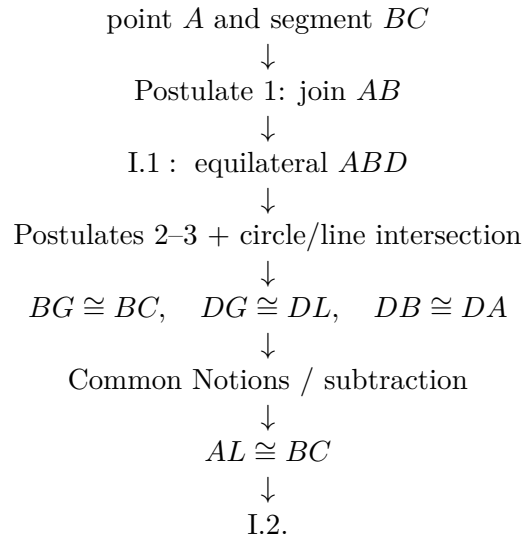
The metric conclusion is purely synthetic:

$$AD \cong BC.$$

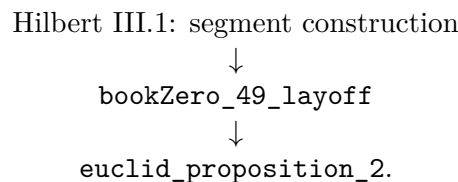
No numerical length function, coordinates, or metric arithmetic is used.

11 Dependency Summary

The classical dependency can be summarized as



The formal dependency is much shorter:



No new geometric axiom is introduced locally by `Proposition02.lean`. The Book I theorem is a thin interface over a foundational congruence operation.

12 Conclusion

Proposition I.2 is one of the clearest early examples of the difference between Euclid’s historical construction order and the Hilbert foundation. For Euclid, transporting a length to a prescribed endpoint requires the full auxiliary mechanism of I.1, produced straight lines, two circles, and subtraction of equal segments. In the current formal development the same operation has already become Hilbert III.1.

The audit therefore identifies three distinct layers:

Euclid’s derived construction, Book Zero layoff interface, Hilbert III.1 primitive.

The public Book I theorem is short not because the geometry has been ignored, but because the relevant construction has been moved to the axiomatic foundation. The unused source-segment nondegeneracy hypothesis also provides a concrete API-cleanliness point for possible later refactoring.